SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

STUDENT NAME : BALAJI A

REGISTER NO : 1911260245

DEPARTMENT : B.TECH COMPUTER SCIENCE ENGINEERING

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FACULTY NAME : KUMARAN G

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1. Given Traceroute Data

A traceroute from the plant to the head-office server (400 km away) gave the values below. Each hop has three probes; the times are round-trip times (RTT) in milliseconds. The values are used exactly as given in the assignment.

Hop	IP address	Probe 1 (ms)	Probe 2 (ms)	Probe 3 (ms)
1	10.10.1.1	0.6	0.5	0.6
2	100.64.0.1	3.1	2.9	3.4
3	172.31.4.9	4.0	**38.2**	4.2
4	103.21.8.17	11.8	11.5	12.1
5	* (no reply)	*	*	*
6	10.200.0.10	12.6	12.4	12.9

2. Propagation Delay Check

Propagation delay is the time a signal needs to travel through the fibre. Given: distance d = 400 km and signal speed in fibre $v = 2 \times 10^8$ m/s.

$$d = 400 \text{ km} = 400 \times 1{,}000 \text{ m} = 400{,}000 \text{ m}$$

$$Tp = d / v$$

$$Tp = 400{,}000 \text{ m} \div (2 \times 10^8 \text{ m/s}) = 0.002 \text{ s} = \mathbf{2 \text{ ms}}$$

This **2 ms is the one-way propagation delay**. Traceroute measures round-trip time, so the ideal round-trip propagation component is:

$$RTT(\text{ideal}) = 2 \times Tp = 2 \times 2 \text{ ms} = \mathbf{4 \text{ ms}}$$

The measured RTT at the final hop (Hop 6) is approximately 12.6 ms. The share explained by propagation is:

$$(4 \text{ ms} \div 12.6 \text{ ms}) \times 100 \approx \mathbf{31.75\ \%}$$

$$\text{Remaining delay} = 12.6 \text{ ms} - 4 \text{ ms} = 8.6 \text{ ms} (\approx 68.25\ \%)$$

This is an approximate comparison. Propagation delay explains about one third of the measured 12.6 ms. The remaining 8.6 ms can include router processing, queueing, access-network delays and other parts of the path. It cannot be assigned to any one specific router from this data.

3. Analysis of Unusual Traceroute Results

3.1 Why does Hop 3 show 38.2 ms?

Two probes at Hop 3 are about 4 ms (4.0 ms and 4.2 ms), but one probe is unusually high at 38.2 ms, about 9 times the normal value ($38.2 \div 4.1 \approx 9.3$). Possible causes are temporary congestion, queueing, or a delayed ICMP response at that router. One high probe does **not** automatically prove the router is faulty: a router may forward normal traffic efficiently while being slow to generate its own ICMP replies. The two normal probes show that the path itself is behaving normally at that moment.

3.2 Why does Hop 5 show * * *?

Stars mean traceroute received no response before the timeout. The router at Hop 5 may block or rate-limit ICMP time-exceeded messages while still forwarding normal traffic. Because Hop 6 replies (12.6, 12.4 and 12.9 ms), packets are clearly passing through Hop 5, so the stars are **not** proof of a complete path failure.

3.3 Which hop adds the most delay?

Traceroute RTTs are cumulative: each hop's time includes the whole path from the source to that hop and back. The average of the three probes at each responding hop is:

Hop	Calculation (ms)	Average RTT (ms)	Comment
1	(0.6 + 0.5 + 0.6) ÷ 3	0.57	Local gateway
2	(3.1 + 2.9 + 3.4) ÷ 3	3.13	Stable
3	(4.0 + 38.2 + 4.2) ÷ 3	15.47	Distorted by the 38.2 ms outlier
3 (normal probes)	(4.0 + 4.2) ÷ 2	4.10	Outlier excluded
4	(11.8 + 11.5 + 12.1) ÷ 3	11.80	Stable
6	(12.6 + 12.4 + 12.9) ÷ 3	12.63	Stable (Hop 5 gave no reply)

Because Hop 3's average (15.47 ms) is distorted by one outlier, the normal value (4.1 ms) is used to find the increase between consecutive hops:

Step	Calculation (ms)	Increase (ms)
Hop 1 → Hop 2	3.13 − 0.57	2.57
Hop 2 → Hop 3 (normal)	4.10 − 3.13	0.97
Hop 3 (normal) → Hop 4	**11.80 − 4.10**	**7.70**
Hop 4 → Hop 6	12.63 − 11.80	0.83

The largest apparent consistent increase is about **7.7 ms between Hop 3's normal responses and Hop 4**. This step is larger than the 4 ms ideal round-trip propagation time, so it may include a long-distance section plus equipment delay. However, differences between consecutive hops are only approximate indicators: they cannot conclusively identify the exact delay added by one router, because each hop's reply can take a different return path and ICMP replies are handled at lower priority.

4. My Own Windows Traceroute

tracert -d google.com

Part	Meaning
tracert	Windows command that traces the route (list of routers) to a destination.
-d	Prevents DNS name resolution, so IP addresses are shown instead of host names (faster output).
google.com	Destination host name.

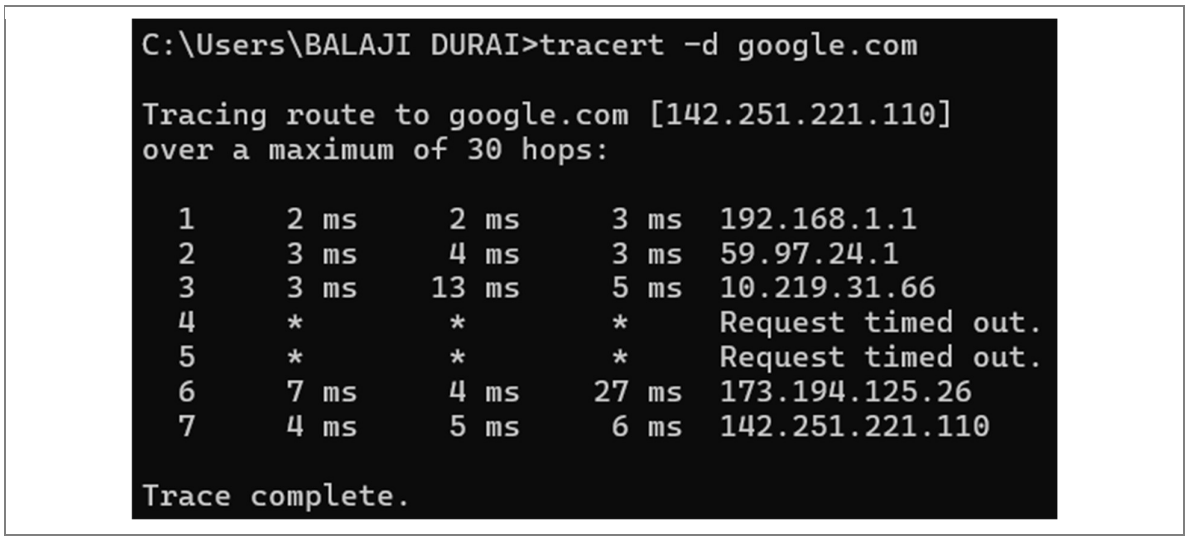


Figure 1: Windows traceroute output.

Results table. Copy the values that are visible in Figure 1 (one row per hop; add rows as needed). Do not use the assignment's sample values.

Hop	IP address	Probe 1	Probe 2	Probe 3	Observation
1	192.168.1.1	2	2	3	Normal response
2	59.97.24.1	3	4	3	Normal response
3	10.219.31.66	3	13	5	Normal response
4	--	--	--	--	Request timed out
5	--	--	--	--	Request timed out
6	173.194.125.26	7	4	27	Normal response
7	142.251.221.110	4	5	6	Reached destination

5. My Own Windows Ping Test

ping -n 10 google.com

Part	Meaning
ping	Tests reachability of a host by sending ICMP echo requests and waiting for echo replies.
-n 10	Sends ten echo requests (the default is four).
Response time	Measured in milliseconds (ms) for each request–reply pair, i.e. the RTT.
Packet loss	Percentage of probes that did not receive a reply.

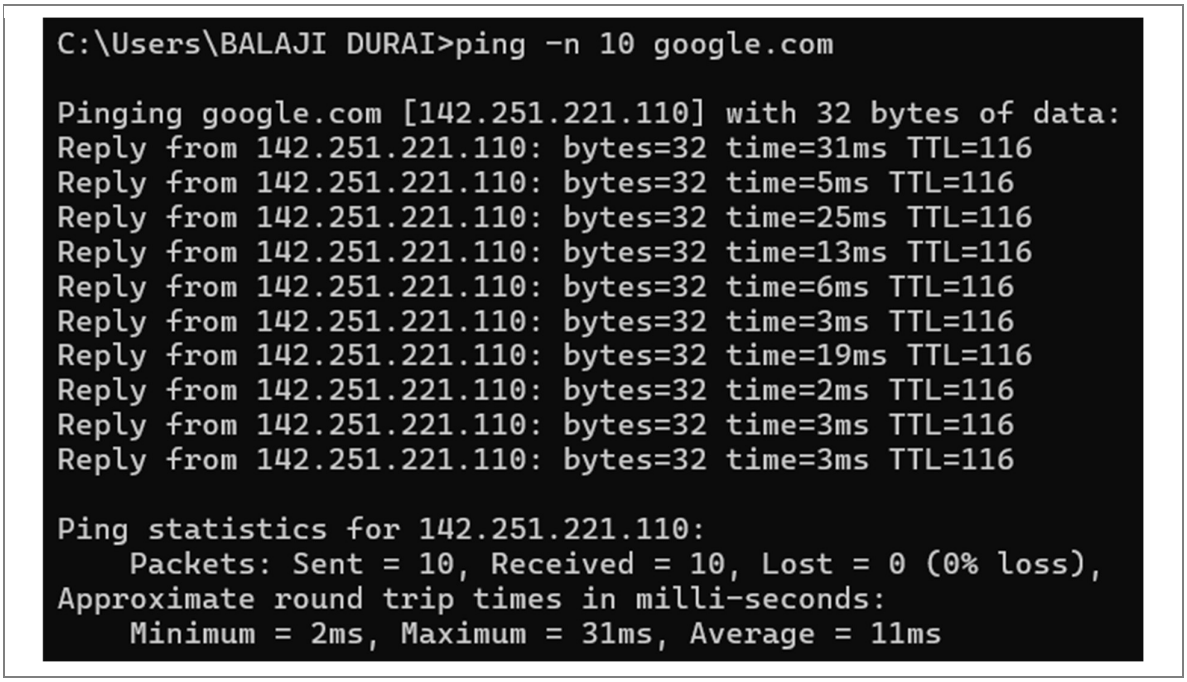


Figure 2: Windows ping test output.

Measurement table. Fill in from Figure 2 only. The formulas are:

$$\text{Packets lost} = \text{Packets sent} - \text{Packets received}$$
$$= 10 - 10$$
$$= 0$$
$$\text{Packet loss (\%)} = (\text{Packets lost} \div \text{Packets sent}) \times 100$$
$$= (0 \div 10) \times 100$$
$$= 0\ \%$$

Measurement	Value (from my screenshot)
Packets sent	10
Packets received	10
Packets lost	0
Packet loss percentage	0
Minimum RTT	2 ms
Maximum RTT	31 ms
Average RTT	11 ms

Meaning of the results. Packet loss of 0 % means every probe was answered; any loss indicates drops on the path or a host that ignores ICMP. The average RTT shows the typical delay, and the gap between the minimum and maximum RTT shows how much the delay varies (jitter). [After inserting Figure 2, add one sentence stating your own result, for example your loss percentage and average RTT.]

6. Wireshark Packet Analysis

Wireshark is a packet analyser. It captures packets passing through a network interface and shows the header fields of every protocol layer in each packet, so a problem can be traced to a specific layer.

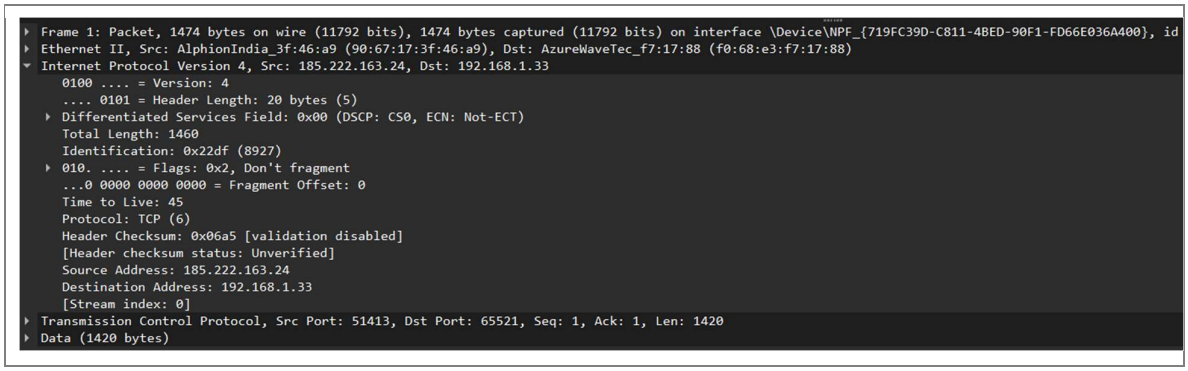


Figure 3: Wireshark packet capture and protocol-layer analysis.

Protocol table. In the third column, write what is visible in your Figure 3. If your selected packet is ICMP (for example a ping), ICMP is a network-layer protocol carried directly in IPv4; do not label it TCP or UDP, and the Transport row then reads "Not used (ICMP)".

Layer	Protocol	Information identified (from my screenshot)	Purpose
Data link	Ethernet II	[source MAC] → [destination MAC]	Delivers the frame to the next device on the same local network using MAC addresses.
Network	IPv4 (or ICMP if the packet is a ping)	[source IP] → [destination IP]	Addresses the packet end to end and routes it between networks.
Transport	TCP	[source port] → [destination port]; [Seq]; [Ack]	Identifies the application port and provides reliable, ordered delivery using sequence and acknowledgement numbers.
Application	[protocol, if visible, e.g. HTTP/TLS]	[application data, if visible]	Carries the actual user or application information.

How the packet travels through the layers. The sender's application creates data. The transport layer adds a TCP header (ports, sequence and acknowledgement numbers) to form a segment; the network layer adds an IPv4 header (source and destination IP) to form a packet; the data-link layer adds an Ethernet header and trailer (MAC addresses, FCS) to form a frame, which is sent as bits. At each router the Ethernet header is replaced for the next link while the IP addresses stay the same. The receiver removes the headers in the reverse order, from Ethernet up to the application.

7. Alternative Concept Analysis: Continuous Monitoring versus Ping/Traceroute

SNMP counters collect interface counters and device statistics (traffic, errors, discards) that can be polled continuously. **NetFlow** records traffic-flow information (source, destination, volume), showing traffic patterns and heavy senders. **Ping** tests reachability, RTT and packet loss at the moment of testing. **Traceroute** shows the route and where RTT changes or replies stop.

Parameter	SNMP counters	NetFlow	Ping / Traceroute
Data collected	Interface counters: bytes, errors, discards, device statistics	Flow records: source/destination, ports, volume	RTT, packet loss, list of hops
Mode	Continuous polling or configured monitoring	Continuous export from router	Occasional, manual snapshot
Temporary latency spikes	Weak: counters averaged over the polling interval hide short spikes	Weak: shows volume, not delay	Detects a spike only if it occurs during the test (as with the 38.2 ms probe)
Persistent interface errors	**Strong:** error and discard counters rise and show trends	Not designed for this	Cannot see error counters
Traffic congestion	Strong: utilisation and queue drops over time	**Strong:** shows which flows use the bandwidth	Indirect: RTT rises during the test
Non-responsive intermediate router	Can show an interface or device down if polled	Shows whether flows still pass, if that router exports flows	**Shows it immediately** (stars), but cannot tell whether it is a fault or ICMP blocking
Historical evidence and trends	Yes	Yes	No (each test is a snapshot)
Detection speed	Depends on polling interval and alert settings	Depends on export and collector settings	Immediate, but only when run

7.1 Which problems in the supplied traceroute would each method detect?

Problem in the traceroute	SNMP counters	NetFlow	Ping / Traceroute
Hop 3 single 38.2 ms probe	Probably not: it affects one ICMP reply, not interface counters	No	Seen only because it happened during the run; one more run may not repeat it
Hop 5 stars	Only if the device or interface is actually down	Shows whether traffic still flows through that path	Seen immediately in the run, but cause is unclear
7.7 ms step from Hop 3 to Hop 4	Would show congestion or drops on the link if they exist	Would show heavy flows using the link	Locates where RTT rises, but only approximately

Continuous monitoring (SNMP and NetFlow) provides trends and historical evidence and can raise alerts without anyone running a test. Occasional ping and traceroute tests give a quick but limited snapshot. Detection speed in all cases depends on the monitoring interval and configuration.

8. Conclusion

- The expected one-way propagation delay over 400 km of fibre is **2 ms**, so the ideal round-trip propagation component is **4 ms**.
- The measured final-hop RTT is about **12.6 ms**, so propagation explains roughly **31.75 %**; the remaining 8.6 ms is processing, queueing and other path components.
- The 38.2 ms value at Hop 3 is an **outlier**, not definite proof of a fault.
- The stars at Hop 5 mean missing traceroute responses, not necessarily a failed router, because Hop 6 replies.
- The largest consistent apparent RTT increase (about **7.7 ms**) is between Hop 3's normal responses and Hop 4; this is only an approximate indicator.
- Ping, traceroute and Wireshark give complementary troubleshooting evidence, while SNMP and NetFlow provide more continuous monitoring information and history.